

Due date: September 8th at 12:00 pm

A soft copy must be uploaded on Gradescope. No late homework will be accepted. Please make sure your Name and Student ID Number are in the top-right hand corner of the page. Number all pages in the bottom-right hand corner. Please describe your thought process and state all assumptions made for each question. For numerical answers, place a box around your final answer and remember units.

Problem 1 (5 points)

Control Temp

Electric kettles are used to boil water rapidly via Joule heating. More sophisticated kettles are able to control the temperature of water instead of requiring that the water boil to open the heating element circuit. For such a kettle, identify the controlled variable (CV), manipulated variable (MV), and any possible disturbance variable (DV). Then, draw a block diagram for the feedback control system. Include the sensor, controller and the actuator.

Problem 2 (B. Wayne Bequette – Edition 2, Chapter 1, Student Exercise 13) - (5 points)

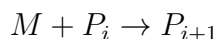
An aeration basin (a very large “vessel”) is used to continually treat a wastewater stream aerobically. A number of mechanical agitators at the surface can be used to “pump” air into the basin; these can be independently turned on or off to change the level of dissolved oxygen in the basin. The microbes that “eat” the waste function best in a small pH range. Acid and base stream flowrates into the basin can be manipulated. An analytical device that can measure the concentration of waste material in terms of BOD (biochemical oxygen demand) is placed in the outlet stream of the basin. The liquid level in the basin is relatively constant since the effluent flows over a weir. In this process, we desire to maintain the pH and BOD at some target levels. Identify the feedback control loops in this system. For each control loop, identify the controlled variable (CV), manipulated variable (MV), and disturbance variables (DV).

Problem 3 (15 points)

Control 2: Oxygen dissolved

A CSTR can be used in radical polymerization processes. Consider an isothermal CSTR that receives feed of solvated fresh monomer M with flow rate q_{in} and monomer concentration $C_{M_{in}}$. A valve is placed on the inlet stream to the CSTR. The CSTR is not constant volume since its outlet flow varies in time as $q_{out}(t) = \alpha\sqrt{h(t)}$, where α is a constant and $h(t)$ is the liquid level in the tank.

In a propagation reaction, a unit of monomer M can react with a polymer chain of length i in an elementary step:



For each reaction, the rate constant is the same, such that the rate of disappearance of M can be written as follows:

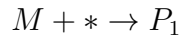
$$\begin{aligned} r_M &= kC_M C_{P_1} + kC_M C_{P_2} + kC_M C_{P_3} + \dots \\ &= kC_M \sum_{i=2}^{\infty} C_{P_i} \end{aligned}$$

For convenience, we define the total polymer concentration as:¹

$$\gamma(t) = \sum_{i=2}^{\infty} C_{P_i}$$

¹This quantity is directly related to the expectation of the final polymer chain length.

The radical chain reaction is started with an initiation reaction:



which has the following rate law:

$$r_{initiation} = k_s C_M$$

The reaction has no termination step, i.e., all polymer species are reactive.

- (a) List the potential manipulated variable(s), controlled variable(s), and disturbance variable(s) for this process.
- (b) Derive the model that describes the dynamics of the liquid level in the tank ($h(t)$), the concentration of monomer ($C_M(t)$), and the sum of polymer concentrations ($\gamma(t)$) as a set of three coupled ODEs. Clearly state all your modeling assumptions. *Hint: $\gamma(t)$ is not not affected by the propagation reaction and is not present in the feed.*

Control Temp: CV, MV, DV

feed back block diagram

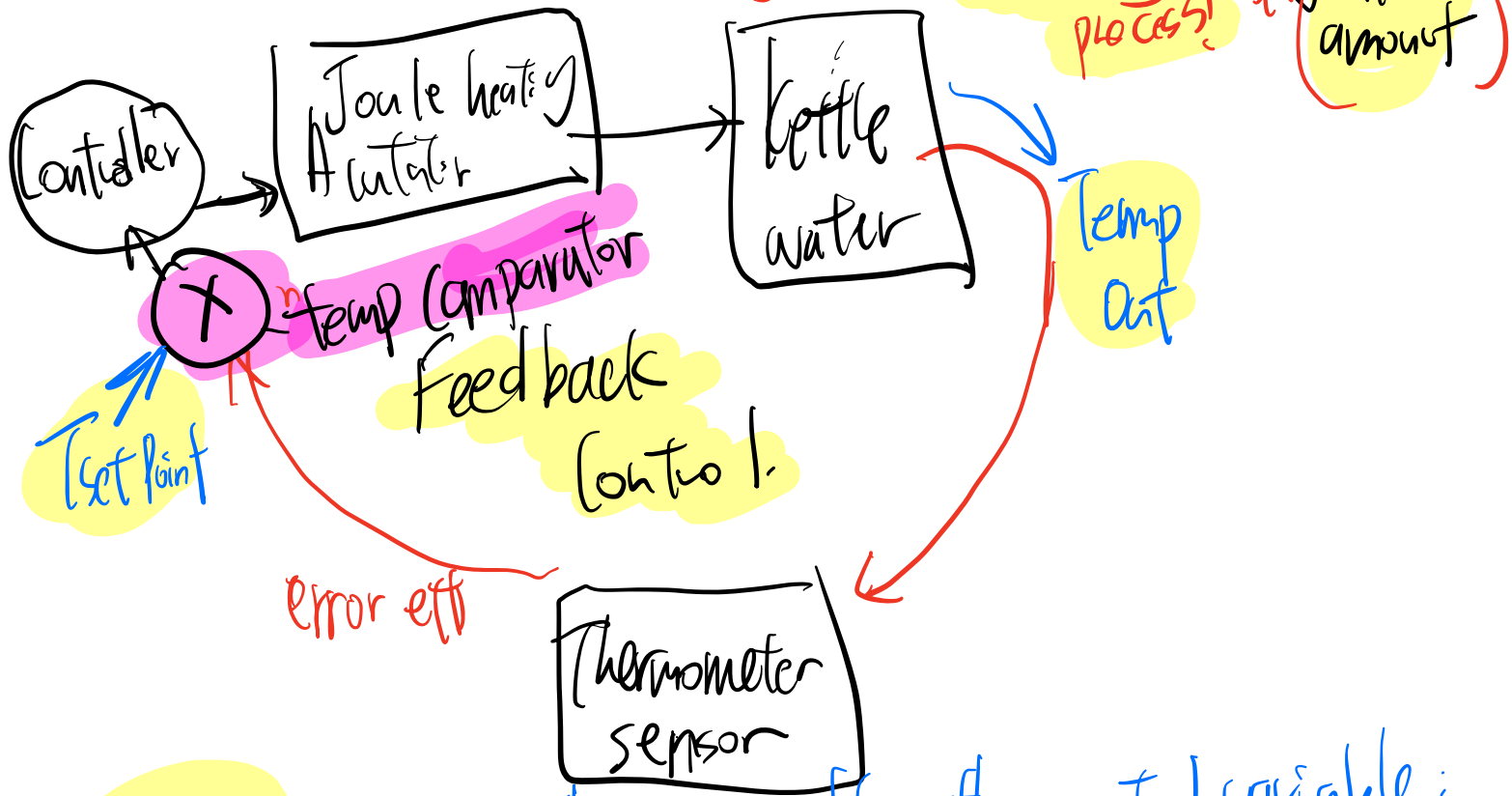
CV = Temperature

MV = Heating Power P or Q $P = I \cdot V$
 (what can I do to control temp)
 (applied to water)

DV = Total temp, Heater Temperature, Surrounding temperature
 (what affects temp but I can't control?)
 Adding water midway the boiling process!
 heat loss (Water amount)



kettle



Feed Back = measure the error after the control variable:
 temperature changes.

$$\text{error}(t) - e(t) = T_{sp}(t) - \text{measure}(t)$$



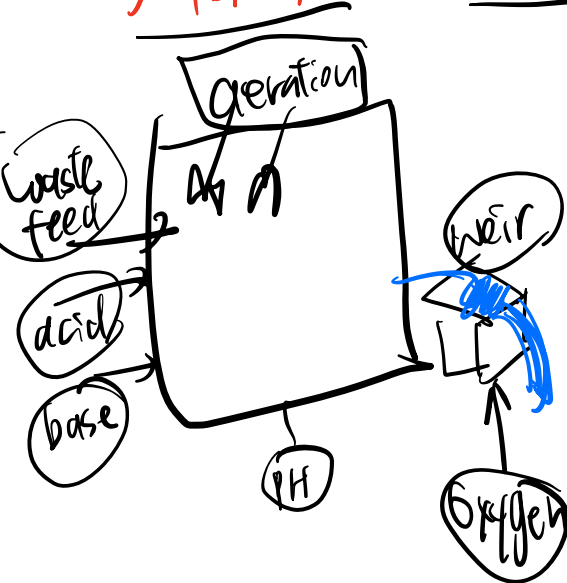
2 feedback control needed! (A)

A) for PH B) for BOD

PH control

CV: PH, MV: base/acid flow rate

DV: waste water composition (Influent PH, Influent flow rate), waste concentration change, buffer present in feed!



BOD control

CV: BOD, MV: number of agitators, aeration rate

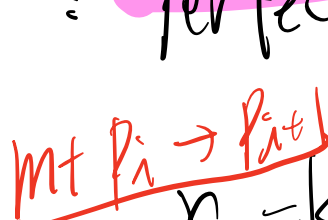
DV: wastewater temperature, microbe health, waste feed oxygen, disolve

3) Continuous Stirred-Tank Reactor (CSTR) Radical Polymerization Process.

key assumption: Perfect mixed CSTR

Isothermal operation

(kinetics constants are



$$r_m = k_m C_m C_{P_1} + \dots$$

$$k_m C_m \sum_{i=2}^{\infty} C_{P_i}$$

rate of monomer disappear

a) $W: h(t)$ height of liquid level
 $C_m(t)$ monomer concentration
 $\gamma(t)$ polymer chain conc

$DV = C_{M \text{ in}}$ Incoming monomer concentration

$MV: q_{\text{in}}$ Inlet flow rate

A valve is placed on the inlet stream

Assumptions

- b) Outlet conc = reactor conc
- liquid density is constant
 - cross area constant $V = A \cdot h$
 - Isothermal, k, k_s constant

Sum of
 of
 polymer conc
 not affected
 by propagation
 not in first feed

$$Q_{\text{out}} = \alpha \sqrt{h}$$

$$\frac{dh}{dt}, \quad \frac{dC_m}{dt}, \quad \frac{d\gamma}{dt}$$

ODEI

$$\frac{dV}{dt} = q_{in} - q_{out} = A \cdot \frac{dh}{dt} = q_{in} - \alpha \sqrt{h}$$

$$\frac{dh}{dt} = \frac{q_{in} - \alpha \sqrt{h}}{A}$$

- level rise if $q_{in} > q_{out}$

- steady state $q_{in} = q_{out}$

ODEU

Accu = In - Out - Reaction Consumption

$$\frac{d(C_m \cdot V)}{dt} = \text{Monomer Number change within time accumulation}$$

$$= V \cdot \frac{dC_m}{dt} + C_m \cdot \frac{dV}{dt}$$

$$I_n = q_{in} \cdot C_{m,in}$$

Molar flow

$$O_{out} = q_{out} C_{out} = \alpha \sqrt{h} \cdot C_m$$

propagation Consumed ($M + P_n \rightarrow P_{n+1}$)

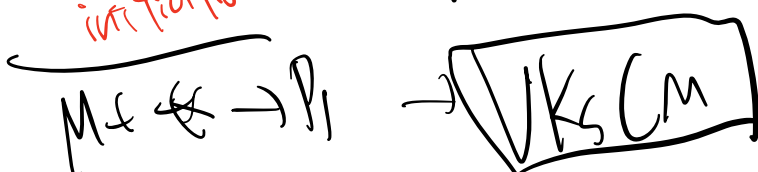
Volume x Consumption

$$= V \cdot K \cdot C_m = \delta$$

δ = polymer conc

Consumed initiation

$$R_{init} = k_i C_m$$



$$\frac{d(m \cdot V)}{dt} = q_{in} C_{min} - q_{out} C_m - V K \cdot C_m \cdot \gamma - V K_B C_m$$

For monomer Concentration (C_m)

$$V \frac{dC_m}{dt} + \cancel{C_m} (q_{in} - \cancel{q_{out}}) = q_{in} \underline{C_{m,in}} - \cancel{\frac{q_{out}}{C_{m,out}}}$$

$$= A h \cdot \frac{dC_m}{dt} = -V \cdot K \cdot C_m \cdot \gamma$$

$$q_{in} (\underline{C_{m,in}} - C_m) - A \cdot h \cdot K \cdot C_m \cdot \gamma$$

$$\frac{dC_m}{dt} = \underbrace{q_{in} (C_{min} - C_m)}_{A \cdot h} - K \cdot C_m \cdot \gamma$$

#

For polymer conc (σ)

no inlet feed

assumption
propagation

don't consume!

$$Acc = \cancel{In} - \cancel{Out} - \cancel{cons} + Gen$$

$$\frac{d\sigma}{dt} = V \cdot k_s \cdot C_m - \cancel{q_{out} \cdot \sigma}$$

$$= V \cdot \frac{d\sigma}{dt} + \sigma \cdot \frac{dV}{dt}$$

$$= V \frac{d\sigma}{dt} + \cancel{\sigma (q_{in} - q_{out})}$$

$$= A \cdot h \cdot \frac{d\sigma}{dt} = A \cdot h \cdot k_s \cdot C_m$$

$$- \sigma q_{in}$$

$$\frac{d\sigma}{dt} = k_s \cdot C_m - \sigma q_{in}$$

#

Polymerization in a Dynamic CSTR

$$\frac{dMA}{dt} \neq 0 \Rightarrow$$

$$V \cdot \frac{dC_A}{dt} = C_{A,in} \cdot Q - C_A \cdot Q - k_r C_A \cdot V$$

$$Acc = in - out - consume$$

$Q = \frac{V}{\tau}$ sec

$Q_{in} \neq Q_{out}$
 $= \alpha \cdot \sqrt{h}$ (not constant outflow)

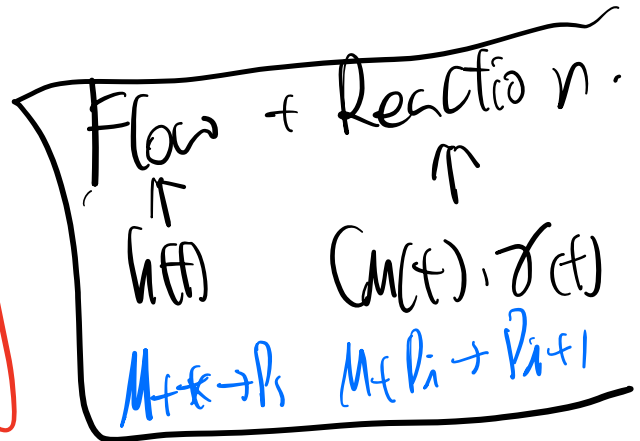
$$V = A \cdot h$$

CSTR is not constant volume

$$\frac{dM}{dt} = (Q_{in} - Q_{out}(t)) \cdot C$$

$$A \cdot \frac{dh}{dt} = (Q_{in} - \alpha \cdot \sqrt{h}) \cdot C$$

Mass balance Equation



$$C_{M,in} \quad Q_{in}$$

$$Q_{out}$$

Perfect Mix

$$C_{M,out} = C_M(t)$$

Assumption

- 1) Perfect mix
- 2) Constant and uniform fluid density
- 3) Constant Area (A)
- 4) Constant $C_{M,in}$

5)

No reaction inlet

Ans

$$h = \frac{Q_{in} - \alpha \sqrt{h}}{A}$$

$$\tau = k_s C_M - \frac{Q_{in}}{A \cdot h} \gamma$$

$$C_M = \frac{Q_{in}}{A \cdot h} (C_{M,in} - C_M) - (k_s + k_r \gamma) C_M$$

Mole on balance Monomer

$$\frac{dC_m(t)}{dt} = q_{in} \cdot C_{min} - \alpha \sqrt{I_h} \cdot C_{max} - h(t) \cdot A \cdot [k \cdot C_m(t) \cdot \gamma + k \cdot C_m(t)]$$

$h(t) \cdot A$
 $\alpha \sqrt{I_h}$

propagation
initiation

Chain rate

$$\rightarrow h(t) \cdot A \cdot \frac{dC_m}{dt} + C_m(t) \cdot \frac{d h(t)}{dt} \cdot A = q_{in} \cdot C_{min} - \alpha \sqrt{I_h} \cdot C_{max} - h(t) \cdot A (k \cdot C_m \cdot \gamma + k \cdot C_m)$$

↳ substitute: $\frac{dh}{dt} = \frac{q_{in} - \alpha \sqrt{I_h}}{A}$

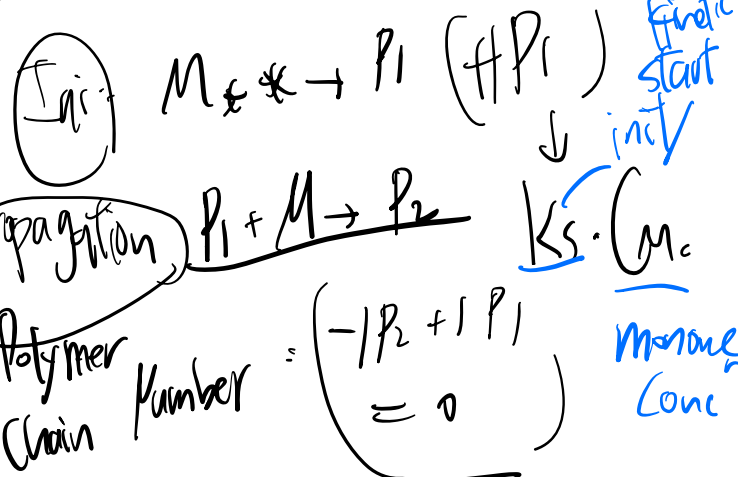
$$\frac{dC_m}{dt} = - \left[C_m \cdot \left(\frac{q_{in} - \alpha \sqrt{I_h}}{A} \right) \right] + q_{in} \cdot C_{min} - \alpha \sqrt{I_h} \cdot C_m - h(t) \cdot A (k \cdot C_m \cdot \gamma + k \cdot C_m)$$

$$h(t) \cdot A \cdot$$

$$= -C_m \cdot q_{in} + C_m \cdot \alpha \sqrt{I_h} + q_{in} \cdot C_{min} - \alpha \sqrt{I_h} \cdot C_m - h(t) \cdot A (k \cdot C_m \cdot \gamma + k \cdot C_m)$$

$$= \frac{q_{in} \cdot C_{min} - C_m}{A \cdot h(t)} - k \cdot C_m \cdot \gamma + k \cdot C_m(t)$$

Mole Balance over Polymer



$$\frac{d(\gamma \cdot V)}{dt} = -(\dot{Q}_{out} \gamma) + V \cdot k_s \cdot C_m$$

out generate (ini)

mole balance eq.

Chain rule

$$= V \cdot \frac{d\gamma}{dt} + \gamma \cdot \frac{dV}{dt}$$

also

$$\frac{dV}{dt} = A \cdot \frac{dh}{dt} = A \cdot \frac{(Q_{in} - \alpha J_h)}{A}$$

$$= V \cdot \frac{d\gamma}{dt} + \gamma (Q_{in} - \alpha J_h)$$

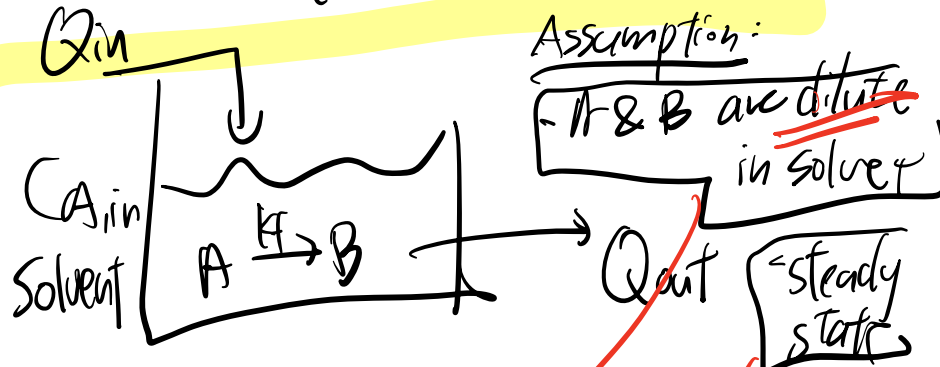
$$\rightarrow \frac{d\gamma}{dt} = \frac{V \cdot k_s \cdot C_m - \cancel{\dot{Q}_{out} \gamma} - \gamma (Q_{in} - \cancel{\alpha J_h})}{V}$$

$$\frac{d\gamma}{dt} = k_s \cdot C_m - \gamma \cdot \frac{Q_{in}}{V}$$

(f)

Continuous Stirred Tank Reactor

CSTR



(Conc Profile) Accum = $I_{in} - Q_{out} + \text{gen} - \text{cons}$

(mass) $\frac{dm(t)}{dt} = m_{in} - m_{out} = \rho_{in} Q_{in} - \rho_{out} Q_{out}$

(Steady state)

$\rho_A \& \rho_B \ll \rho_{\text{solvent}} \rightarrow \rho_{in} = \rho_{out} = \rho_{\text{solvent}}$

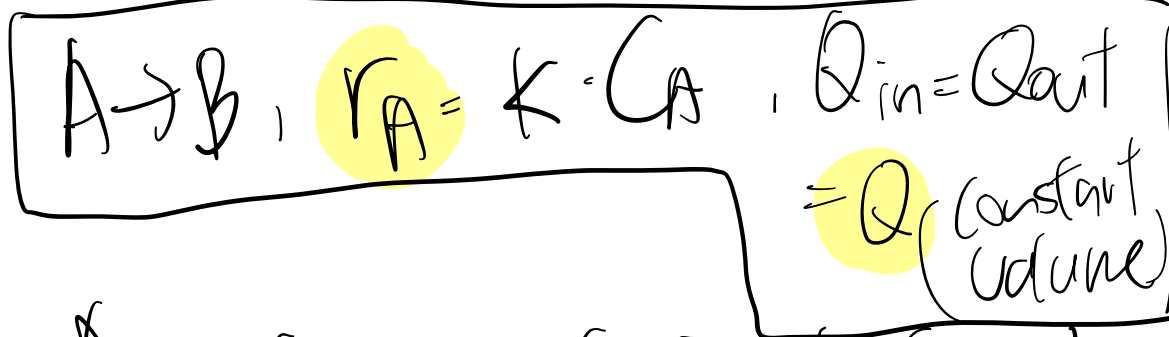
$\rho(t) \approx \rho_{\text{solvent}} \approx \rho_{\text{pure solvent}}$

(A, B are not a function of time)

$$\frac{dC_i \cdot V}{dt} = \underbrace{C_{i,in}}_{in} Q_{in} - \underbrace{C_{i,out}}_{out} Q_{out} + \underbrace{r_i V}_{reaction}$$

for an uniform system

Concentration
in
steady CSTR



$$0 = C_{Ain}Q - C_AQ - k \cdot C_A \cdot V$$

$$\frac{d(C_A \cdot V)}{dt} = 0$$

$$\rightarrow C_{Ain} \cdot Q = C_A (Q + k \cdot V)$$

$$\rightarrow C_A = \frac{C_{Ain} \cdot Q}{Q + k \cdot V}$$

residence
time

$$\tau = \frac{V}{Q} \left(\frac{\frac{m^3}{m^3/s}}{\frac{m^3}{m^3/s}} = s \right)$$

$$= \frac{C_{Ain}}{1 + \frac{k \cdot V}{Q}} = \frac{C_{Ain}}{1 + k \cdot \tau}$$

Conversion

$$X_A = \frac{C_{Ain} - C_{Aout}}{C_{Ain}}$$

$$X_A = \frac{C_{Ain} \cdot Q - C_A \cdot Q}{C_{Ain} \cdot Q} = 1 - \frac{C_A}{C_{Ain}}$$

for 1st order CSTR $\frac{C_A}{C_{Ain}} = \frac{1}{1 + k \cdot \tau}$



